

- **Project : Netflix Movies TV Shows Clustering Unsupervised ML**

- **Tools Used:** Pandas, NumPy, Scikit-learn, Seaborn
- **Industry Context:** The streaming industry, particularly Netflix, has transformed how audiences consume media. With an ever-expanding library of movies and TV shows, there is a growing need for sophisticated data analytics to understand viewer preferences, optimize content offerings, and enhance user experience. As of 2019, Netflix faced challenges in categorizing its vast content effectively and recommended relevant titles to its users.
- **Problem Definition:**
 - Analyze a comprehensive dataset of Netflix movies and TV shows as of 2019.
 - Utilize unsupervised learning techniques to cluster similar content based on textual features.
 - Develop a content-based recommender system that enhances user experience by providing personalized content suggestions.
- **About Data:**
 - **Dataset Overview:** Contains information on 7,787 titles, including 12 columns: show ID, type (Movie/TV Show), title, director, cast, country, date added, release year, rating, duration, genre listings, and descriptions.
- **Data Preprocessing:**
 - **Data Cleaning:** Handled missing values, ensuring the dataset was analysis-ready by addressing null entries in critical columns such as director and cast.
- **To prepare for clustering:**

Text Processing: Cleaned and normalized text fields by removing stop words, punctuation, and converting all text to lowercase.

Vectorization: Employed TF-IDF (Term Frequency-Inverse Document Frequency) to transform textual data into numerical format suitable for clustering.

- **Actionable Insights using Exploratory Data Analysis (EDA) (4 - 5 Insights):**
 - Used PCA that reduced dimensionality and removed the least important variables from the model.
 - Optimize Content Length: Prioritize movies between 80-120 minutes and invest more in single-season TV shows to match viewer preferences.
 - Strategic Content Scheduling: Release emotional, festive, and family-oriented content during peak viewing months (Oct-Jan) to maximize engagement.
 - Diversify Production & Ratings: Increase investment in emerging markets like India and explore more content for underrepresented ratings (e.g., G, TV-Y) while maintaining focus on popular adult-rated content.
 - Targeted Genre Expansion: Continue investing in popular "International Movies" and "Dramas," but also explore expanding into underrepresented genres like "Romantic Movies" to broaden audience appeal.
- **Data Preparation:** The entire dataset was used for clustering, as clustering algorithms typically do not require a train-test split.
- **Model Selection:**
 - Developed a simple baseline model like K-means with a small number of clusters for initial exploration.
 - Experimented with more complex models (Hierarchical Clustering) to capture different shapes of clusters.
- **Model Training:**
 - Optimize model parameters such as the number of clusters in K-means (e.g., using the elbow method or silhouette analysis)
 - PCA was applied to reduce noise and optimize the clustering process.
- **Model Evaluation:**

- Validate the quality and interpretability of clusters by examining cluster centroids, the distribution of features within clusters, and visualizing clusters
- Word clouds can also be used to understand the common terms within each cluster.
- The dendrogram indicated 6 as the optimal number of clusters for this dataset.
- **Feature Importance:** The focus was on defining and understanding the characteristics of each identified cluster. This involves analyzing the common features (genres, directors, cast, keywords from descriptions) that define each cluster.
- **Key Results:**
 - Movie vs. TV Show Distribution: Movies comprised approximately 69% of the dataset, highlighting a strong viewer preference for film content .
 - Optimal Clustering: The elbow method and dendrogram analysis (with the optimal cut at 6) indicated that 6 was the optimal number of clusters for the K-Means algorithm, providing distinct content groupings.
 - Content-Based Clustering: Successfully clustered Netflix content into meaningful groups based on textual and categorical features, enabling a better understanding of content categories and relationships.
 - Marketing Strategies: Insights gleaned from the clusters can inform targeted marketing campaigns aimed at specific demographics.
- **Business Impact:**
 - Data-Driven Content Strategy: Insights from clustering can significantly enhance Netflix's content acquisition strategies, focusing on popular genres and improving content diversity based on identified clusters.
 - Targeted Recommendations: The developed clusters can serve as a foundation for building or enhancing a content-based recommender system, leading to more personalized and accurate content suggestions, thereby improving user engagement and retention.
 - Optimized Marketing: Insights from content clusters can inform targeted marketing campaigns, allowing Netflix to promote content to specific demographics more effectively.
- **Future Scope:**

- Addition of Data: Incorporate new content data periodically to ensure the clustering model remains relevant and accurate.
- Experiment with Algorithms: Explore other clustering algorithms (e.g., topic modeling, neural network-based clustering) for potentially more nuanced groupings.
- Hyperparameter Tuning & Feature Engineering: Further optimize model parameters and explore more sophisticated feature engineering techniques, including advanced text embeddings, to improve cluster quality and interpretability.
- **Expected Interview Questions:**
 - Can you provide an overview of the Netflix Movies & TV Shows Clustering project you worked on?
 - What were the primary objectives of the project, and how did you go about achieving them?
 - Could you explain the dataset you utilized for this project, including its source and the types of information it contained?
 - What preprocessing steps did you undertake to prepare the data for analysis, and why were these steps necessary?
 - How did you implement unsupervised clustering techniques, and what tools did you use?
 - Can you discuss the evaluation metrics you applied to assess the clustering performance?
 - What actionable insights did you derive from the clustering analysis, particularly regarding viewer preferences?
 - What challenges did you encounter during the project, and how did you address them?
 - What specific contributions did you make to the project, and how did they contribute to its overall success?
 - Can you discuss the business impact of your findings and recommendations, particularly regarding content strategy?
 - How did you communicate your findings and recommendations to stakeholders, and what was their response?
 - Looking back, is there anything you would have done differently in approaching this project?
 - How do you see the insights and strategies developed in this project benefiting the streaming industry as a whole?
 - In what ways do you think advancements in data analytics will continue to shape the future of streaming services?

